

Applied Policy Evaluation

Implementation Science, AI, and Modern Causal Inference

Eric Booth

Senior Researcher, Texas 2036

www.Texas2036.org

www.github.com/ericabooth (slides and materials)

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Lecture 1: The Modern Evaluator's Toolkit

1. What does modern policy and program evaluation look like in an applied context?
2. How do we make applied evaluation visible and impactful?
3. How is the evaluator toolkit changing?

Relevant and Impactful Evaluation of Policy and Programs

- Person-centered
- Actionable
- Accessible
- Concise

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- Person-centered
- Actionable
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- Concise
- Transparent
- Supports program scalability
- Promotes sustainability
- Modern tool-embracing: outcomes, measurement, ROI

Classic evaluation

Core principals Planning, Measurement, Analysis, and Feedback

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Relevance focused on externalization/generalization "Does this program or intervention work?"

Timing often post-hoc w/feedback for future implementations and evaluation

Deliverables classic reports, publications, policy briefs, toolkits or dashboards

Emerging, modern evaluation

It redefines some of those elements . . .

Core principles Contextual mapping, Implementation Fidelity, and Adaptation

- Implementation Frameworks (e.g., CFIR, RE-AIM)
- Continuous, Iterative Feedback Loops and Improvements

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Relevance focused on context and mechanism: "How and why does this work *here* or for *this group*? Who is left out or falls through the gaps?"

Timing iterative and real-time; experiments embedded directly into the rollout

Deliverables active diagnostic engines, adaptation / course-correction logs, gap analyses, quantification of uncertainty (*Trustworthiness of results*)

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*Applied evaluation is no longer just an external, 3rd party audit...
it is a strategic partnership in the implementation lifecycle.*

Implementation Science in Practice

If clinical or policy research asks:

"Does this intervention improve outcomes in a controlled environment?" (Effectiveness, Causality)

Implementation Science asks:

"Why do evidence-based practices frequently fail to reach their full potential in real-world settings, and how do we systematically bridge that gap?"

Frameworks:

- **RE-AIM playbook:** Reach, Effectiveness, Adoption, Implementation, Maintenance.
- **Consolidated Framework for Implementation Research (CFIR):** Consideration of site characteristics to explain *why* things work in specific settings; work iteratively with practitioners to understand complex environment and implementation.

... Why does this matter?

Implementation Science in Practice: Why Important?

tl;dr IS transforms evaluation *from a "passive report-card" at the end of a grant cycle* into an **active diagnostic engine that provides real-time data to help** program funders, nonprofit leaders, and state agencies course-correct in real-time.

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Evidence is not self-executing: A policy that works in a controlled pilot often fails when scaled. IS provides the empirical tools to measure *why*... locating the exact friction in leadership buy-in, incentives, or frontline capacity.

Context is a feature, not a bug: Classical evaluation controls for the "setting" as noise; IS investigates the setting (both inner organizational capacity and outer political environment) as a possible driver of success.

Closing the "Know-Do" Gap: moves beyond outcome measurement to provide a rigorous approach :

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Closing the "Know-Do" Gap: moves beyond outcome measurement to provide a rigorous approach :

- **Fidelity Monitoring:** Distinguishing between *program failure* (bad idea) and *implementation failure* (bad execution).
- **Systematic Adaptation:** Modifying a policy for local needs without losing its core "active ingredients."
- **Scale & Sustain:** Planning for permanent integration from Day 1, rather than just surviving a pilot grant cycle.

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What did the other AI LLMs miss or mistake?
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What did the other AI LLMs miss or mistake?
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6. **Gut-check and scrutinize** the work with trusted stakeholders and experts.
Evaluation and research as a profession is about taking responsibility for the truth.



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Embracing AI can certainly be:

- (1) efficient (2) evidence of being a custodian of funder or public dollars
- (3) higher quality (4) a democratizer of sophisticated / expensive methods or tools.

Apparent AI content, especially when it lacks substance or accuracy, may alienate your funders, stakeholders, and audiences. *IS may magnify this risk*

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... When audiences lose trust - it's difficult to earn it back.

Data, Analysis, and AI

(roadmap for next session this afternoon)

- **Questions in focus:** Example: Loss of medicaid coverage at 60 days postpartum.
- **Connections to data:** ICPSR, Accessing APIs
- **Analysis in realtime:** Stata and R as core tools; open-science framework
- **Leveraging AI strategically:** Productization ; Non-transparent/non-reproducible but useful none-the-less; check for error and blindspots
- **Results that matter:** Clarity earns attention; attention enables impact

Data, Analysis, and AI

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The choice in the analysis toolkit is not between binary trade-offs: descriptive vs. inferential, summary vs. causal, formative vs. predictive, etc

Tell stories with analyses that:

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- **Zoom out:** Summarize and tell the story for the outcome of interest
 - **Find the lever:** Pick outcomes that can be augmented!
 - Inspection by sub-population or time period of interest (e.g. cross-tabulations)

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 - Inspection by sub-population or time period of interest (e.g. cross-tabulations)
- **Dig deeper:** consider unpacking patterns that match real world complexity or explore 'what if' scenarios
- **Zoom in:** vignettes of *common stories/profiles* of study population

Lecture 2: From Archive to Action

Technical Walkthrough: Medicaid Expansion & PRAMS

- Sourcing Data from ICPSR (study #114516)
- The "Postpartum Cliff" Policy Shock.
- Staggered Adoption & A Causal Story.
- Policy tool: Difference-in-Difference analysis (via Stata `csdid`)
- 3 Hands-on examples in Stata & R

Key tools to consider

- Data accessible via API (automate download, updates when new data come out)
- Use AI to access and clean/harmonize data (but use cross-checks for data integrity!)
- Carefully check codebooks for recoding, weighting, or data inclusion rules (gut check with experts or existing literature)

Scenario: The Postpartum Cliff

- **Policy:** ACA Medicaid Expansion (Continuous Coverage).
- **The Problem:** Loss of coverage at 60 days postpartum.

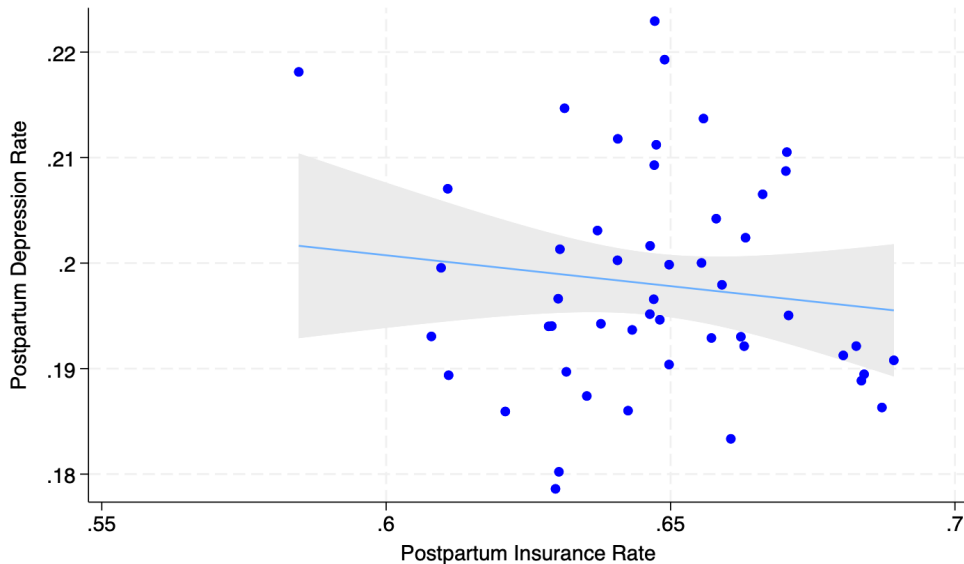


The Context: Baseline Characteristics (2010)

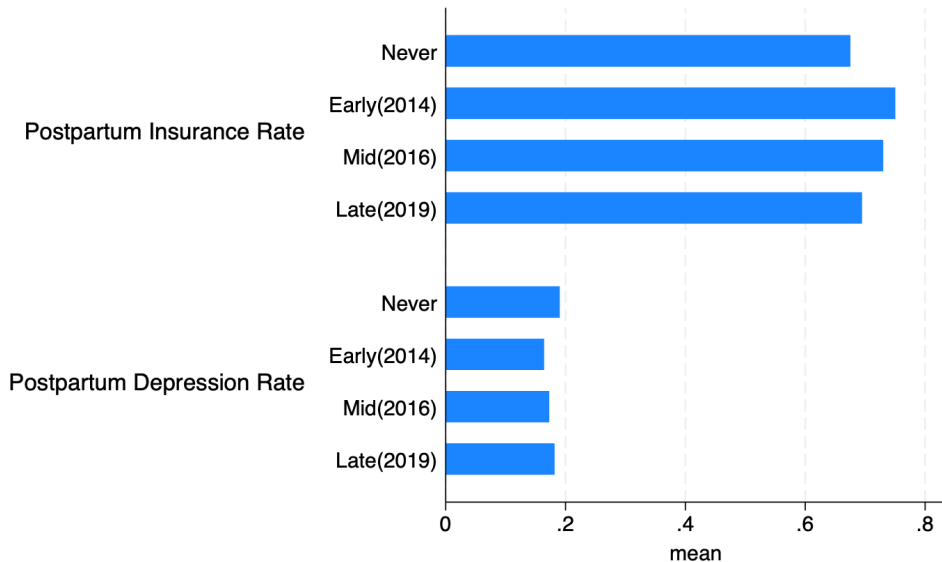
	2010 Avg	2020 Avg	Diff
Postpartum Insurance Rate	64.7	78.0	+13.3%
Postpartum Depression Rate	19.8	15.3	-4.5%

Issue: doesnt account for complexity of the roll-in of the "treatment" policy intervention across time and states

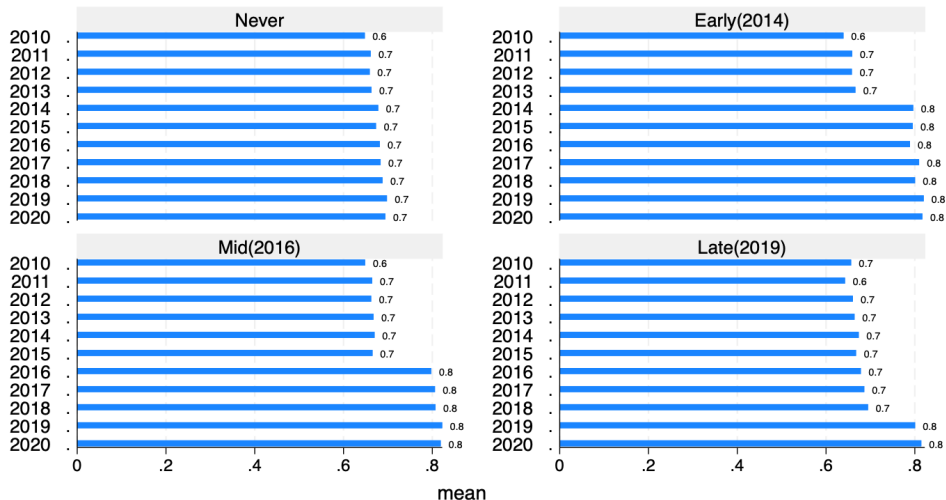
Data exploration



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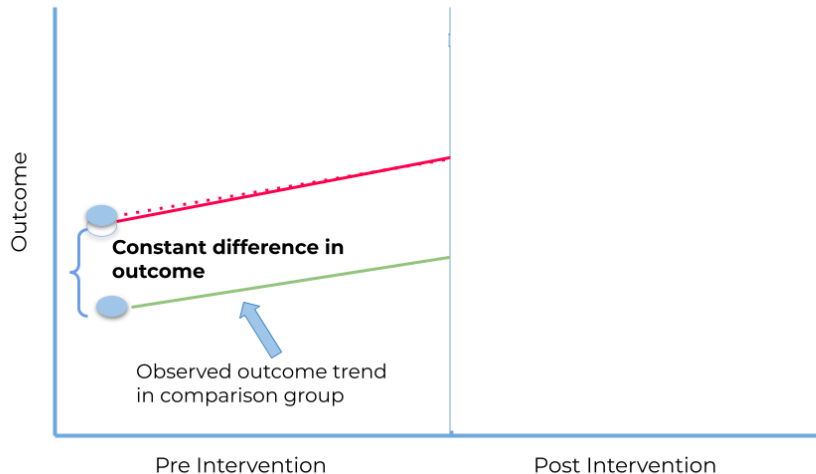
Graphs by Expansion Status

Detecting changes before and after an intervention

We compare how much outcomes changed over time in states that adopted the policy versus not; **the difference between those two changes** is our **intervention effect**

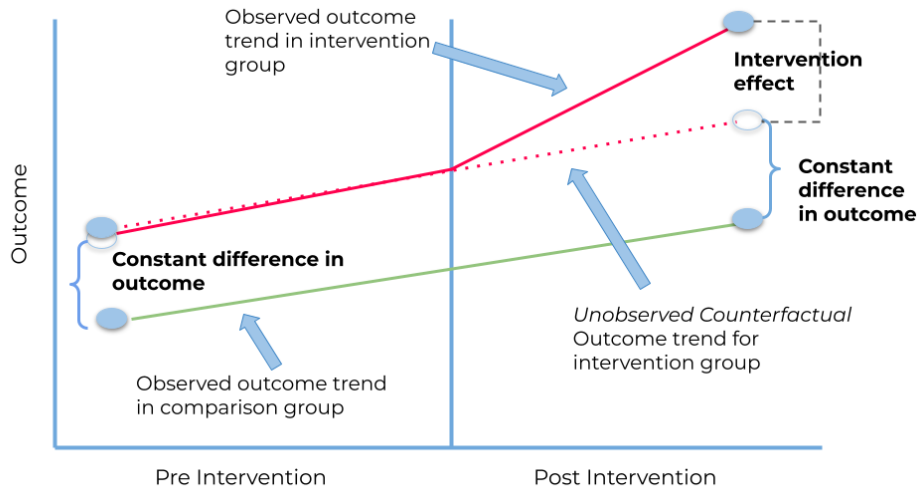
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Implementing Modern DiD

Stata Command: `csdid`

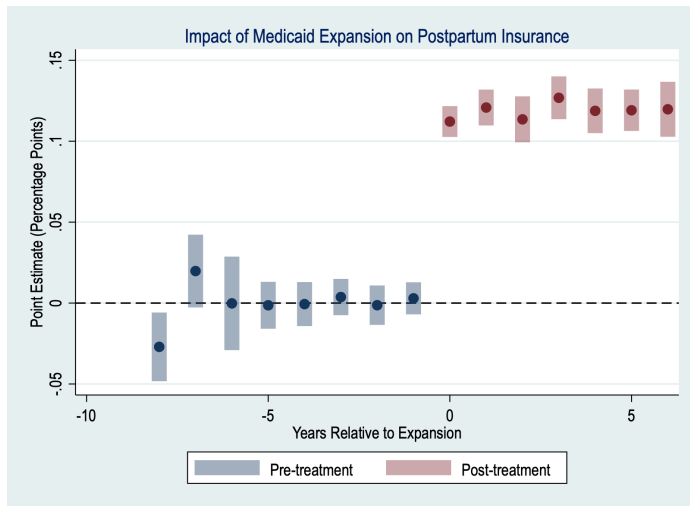
- **ivar:** Panel ID (State).
- **tvar:** Time variable (Year).
- **gvar:** Treatment group (Year of expansion, 0 if never).

```
// DID Aggregate Effects
csdid insured_rate, ivar(state) time(year) gvar(gvar)

    // Simple average
    estat simple
// Group-time average
    estat group

// Diff-in-Diff Study Visual
csdid_plot
```

The Picture: Event Study Plots



*No upward Pre-treatment trend; Post-treatment sustained post-treatment boost.
Each cohort shows a large, immediate, and sustained jump in postpartum insurance rates in the year of expansion*

The Table: Aggregated / Adjusted Effects

Outcome: Insurance Rate	Estimate
Aggregate ATT	0.118*** (0.004)

Aggregated Simple Average Effect Size (ATT) across all groups and times. Aggregate ATT = +11.8 percentage points (SE=0.004, $z=28.16$, $p < 0.001$, 95% CI: 11.0–12.7 pp).

Group-level averages are tightly clustered (10.5 –12.9 pp), indicating consistent effects regardless of when a state expanded.

Synthesis:

Detecting the "So What?" using AI as a thought partner

Translate the model: Package estimates in plain language and visualizations for three audiences: a state legislator, a frontline caseworker, and an affected mother.
Where the stories diverge, you may have a policy lever.

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Translate the **model**: Package estimates in plain language and visualizations for three audiences: a state legislator, a frontline caseworker, and an affected mother. **Where the stories diverge, you may have a policy lever.**

Surface the **knowledge gap**: AI can help unpack where insurance gain *should* produce downstream — birth outcomes, ER utilization, mental health — then flag where your data cannot yet follow. **That gap is your next research question.**

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Preserve the human fingerprint: AI drafts the frame; you supply the texture. Rewrite: **anchor to authentic vignette or stakeholder experience.**

Hands on: Walk through of a few Stata & R examples using ICPSR datasets

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3. TX STAAR performance before/after test redesign in Spring 2023 (Using API & AI to download and merge together data overtime) from Dell Foundation funded [Zelma.ai](#)

Questions?

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eric.a.booth@gmail.com
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